Chapter 9

Bonding and Molecular Structure:

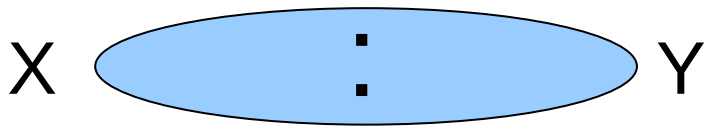
Orbital Hybridization and Molecular Orbitals

Valence Bond Theory

- Bonds are formed when orbitals on atoms overlap
- Overlapping orbitals hold two electrons of opposite spin. Usually, one electron is supplied by each of the two bonded atoms.
- Sigma (σ) and pi (π) bonds arise through orbital overlap.
- *σ bond:* overlap of orbitals along the bond axis
- *π bond:* overlap of orbitals above and below the bond axis.
- *Single bond:* one σ bond
- *Double bond:* one σ bond & one π bond
- *Triple bond:* one σ bond & two π bonds
- **Half-filled atomic orbitals** on bonding atoms overlap to form bonds.
- Hybrid orbitals used to match the electron-pair geometry of the central atom.

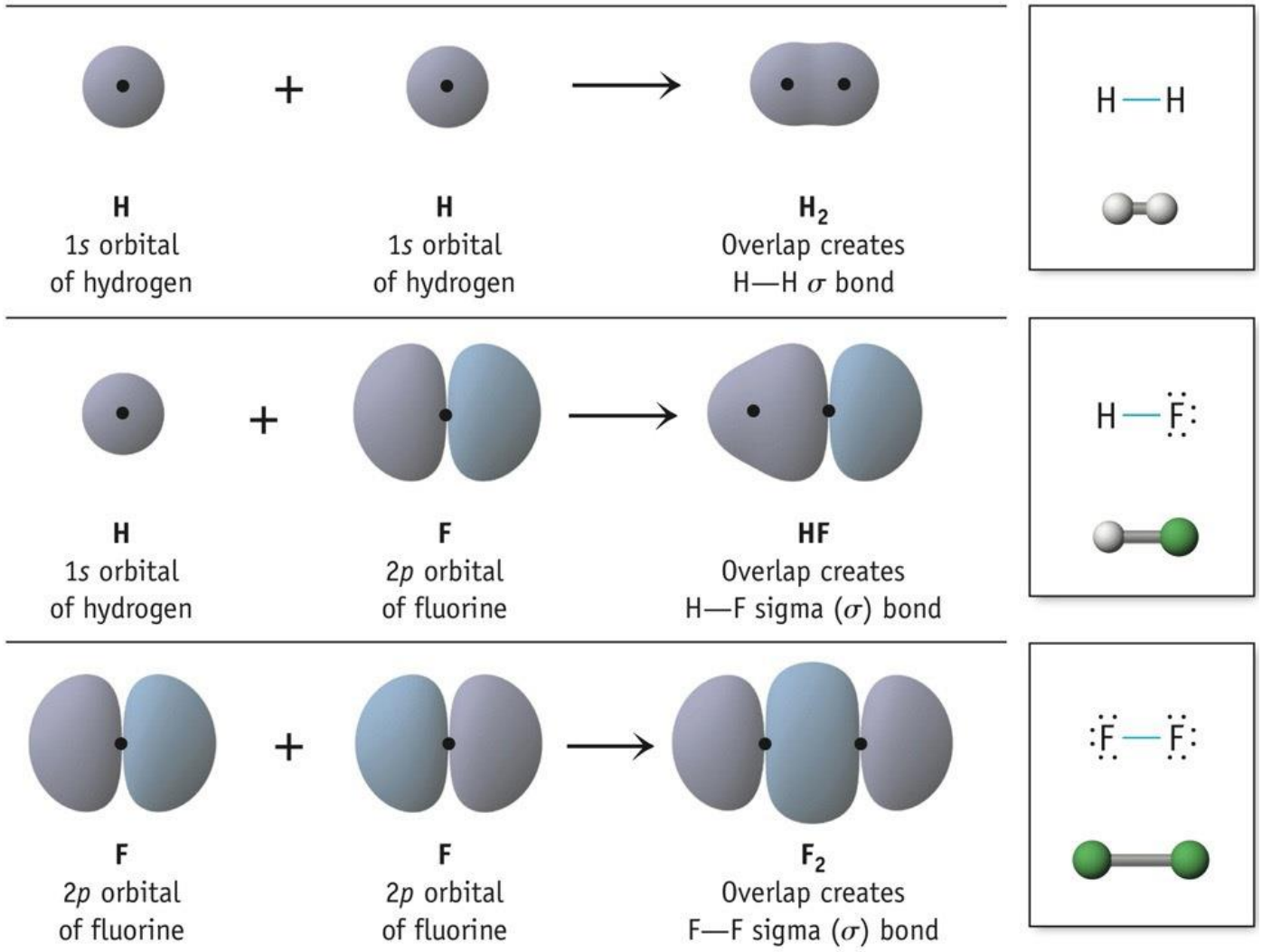
Sigma (σ) Bonding

When bonding occurs along a bond axis, it is referred to as a “sigma” bond: (σ)



The electrons occupy space between the nuclei.

Sigma Bond Formation



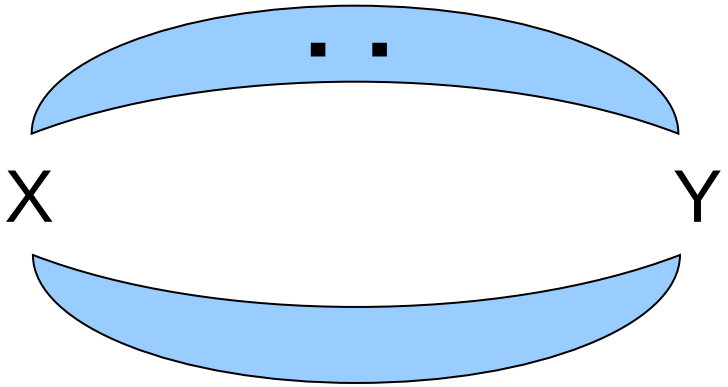
Two *s* orbitals overlap

Overlap of an *s* & *p* orbital

Two *p* orbitals overlap

Pi (π) Bonding

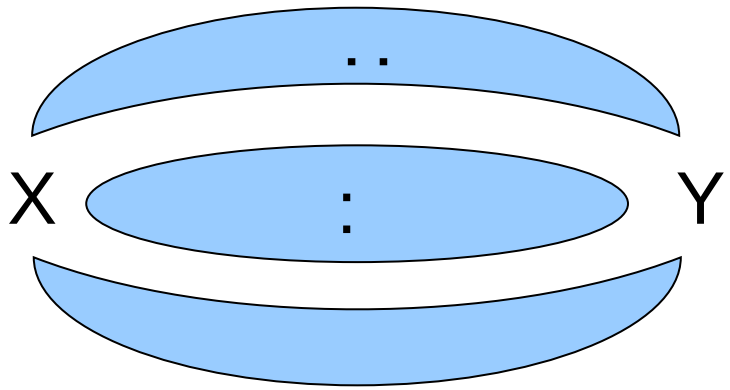
When bonding occurs above and below a bond axis, it is referred to as a “pi” bond: (π)



The electrons occupy space above and below the nuclei.

Sigma (σ) Bonding & Pi (π) Bonding

Sigma (σ) bonding: When bonding occurs along a bond axis, it is referred to as a “sigma” bond.



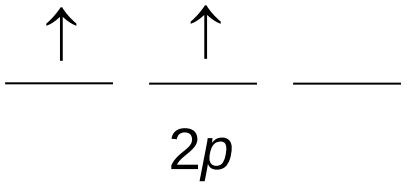
Pi (π) bonding: When bonding occurs above and below a bond axis, it is referred to as a “pi” bond.

A double bond is made up of a σ bond and a π bond.

Hybridization of Atomic Orbitals

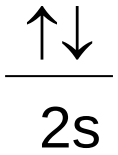
In order to attain the needed geometry, the *atomic orbitals* (AOs) mix or *hybridize* to form new *valence bond* orbitals.

Consider carbon as a central atom in a molecule:



The valence orbitals are the 2*s* & 2*p* orbitals.

There are 4 valence electrons:



carbon

sp^3 Hybridization

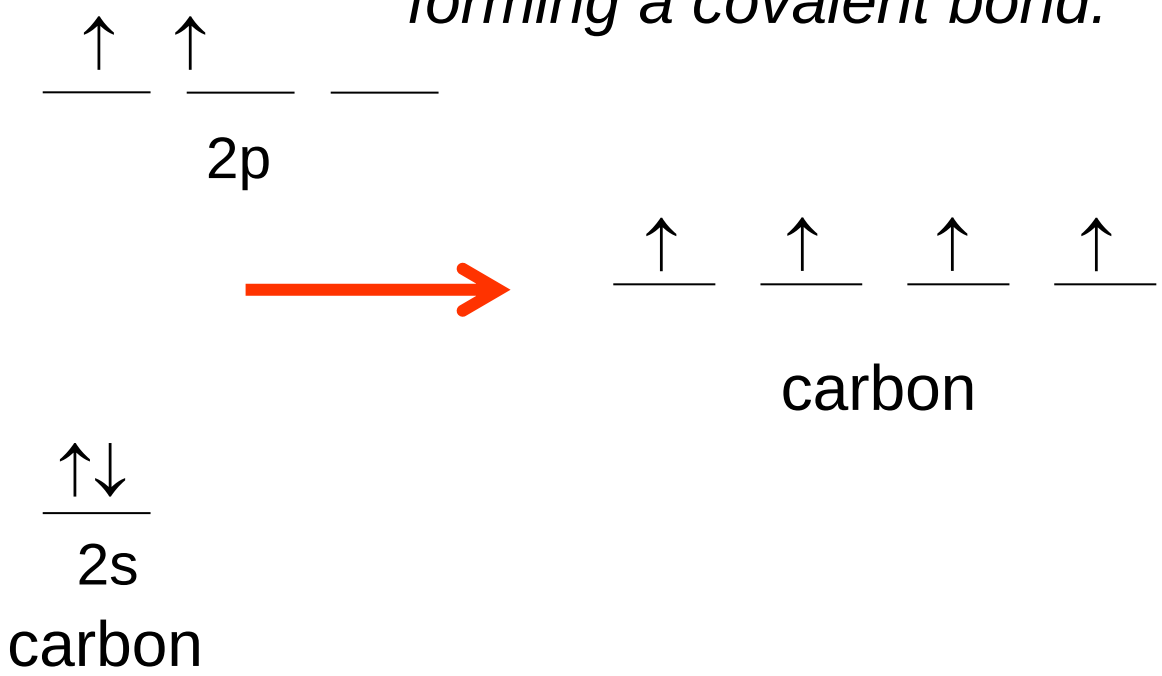
When the central atom in a molecule has a combination of four total sigma (single) bonds and lone pairs, the hybridization at the central atom is sp^3.

$s + p + p + p$
or sp^3 hybridization

sp^3 Hybridization

This hybridization determines the electron-pair geometry for the central atom.

Each half-filled orbital is capable of forming a covalent bond.



The new orbitals are called

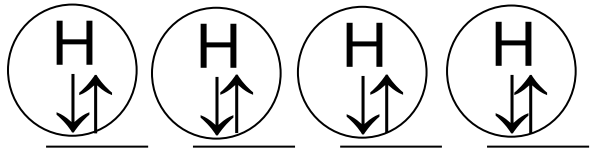
sp^3

Valence bond orbitals

The four valence electrons on carbon fill the orbitals by Hund's rule.

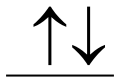
Hybridization of Atomic Orbitals

Bonding in *methane* involves the overlap of the new sp^3 hybrid orbitals in carbon with the 1s orbitals in hydrogen.



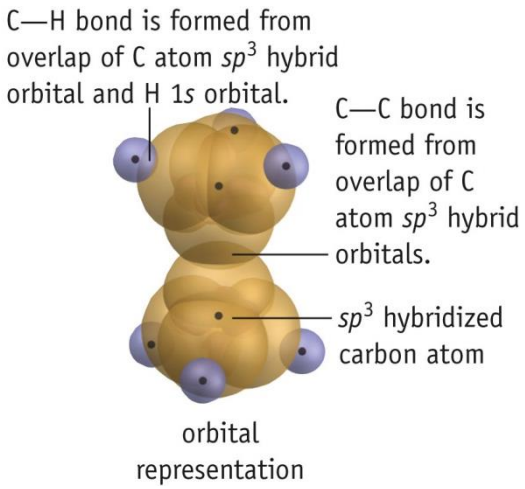
sp^3 hybrid valence bond orbitals

Carbon



Each overlap contains two shared electrons, one from each bonding nuclei.

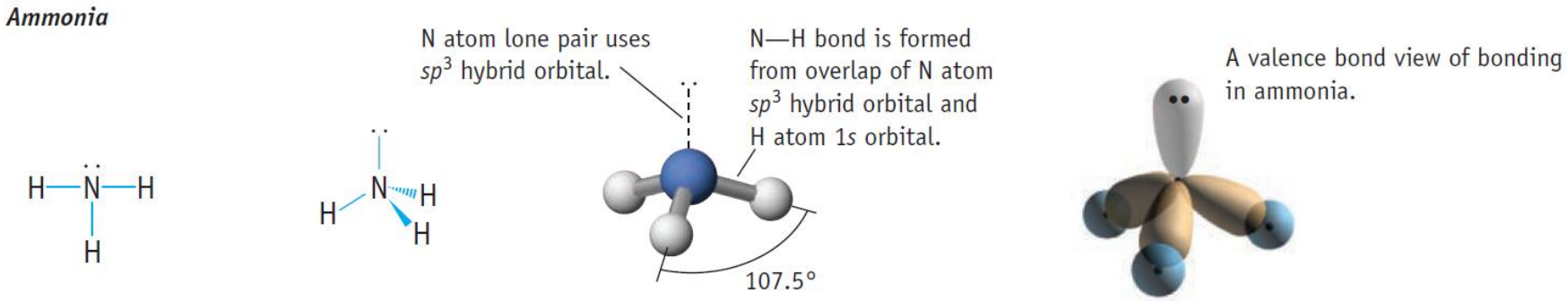
Each overlap results in a σ bond.



sp^3 Hybridization

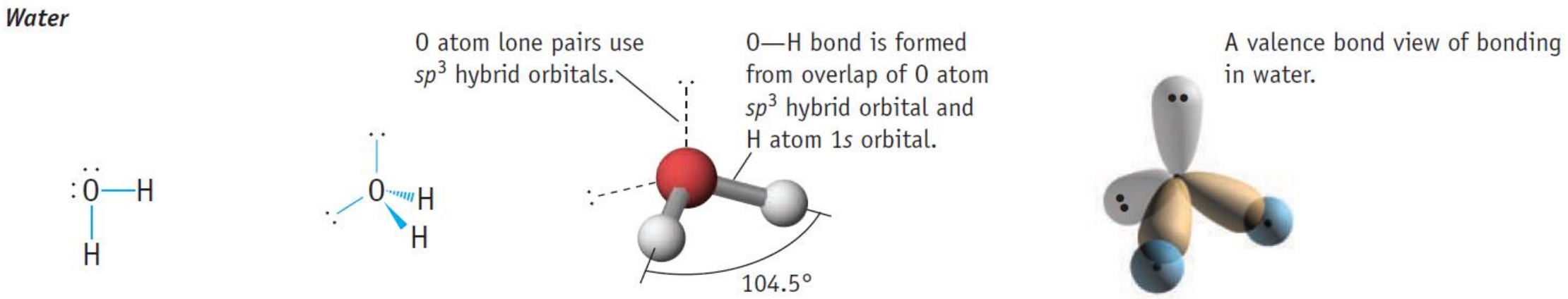
Since there are four sp^3 hybrid orbitals, they must spread out to form a tetrahedron about a central atom to minimize repulsion.

It follows then that a central atom that has a "*tetrahedral*" EPG must have sp^3 hybridization.



sp^3 Hybridization

Water must also bond via sp^3 hybrid valence bond orbitals.



sp^2 Hybridization

When the 2*s* & two of the 2*p* orbitals mix:

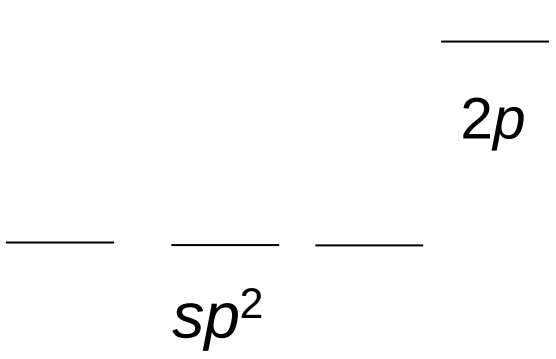
The new orbitals are called sp^2 *Valence bond* orbitals

s + *p* + *p*

or sp^2 hybridization

whenever we mix a certain number of atomic orbitals, we get the same number of hybrid orbitals

The new hybrid obitals all have the same energy:

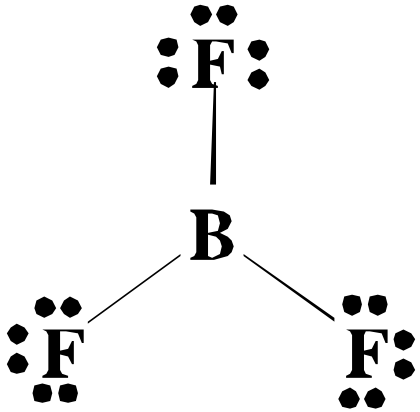


there is one *p* orbital left over

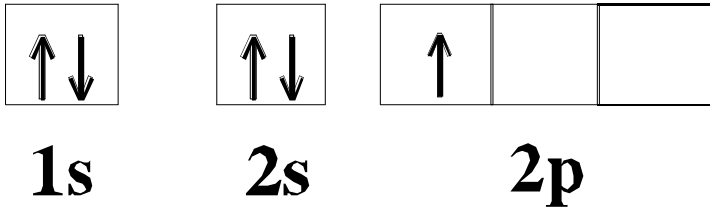
sp^2 Hybridization

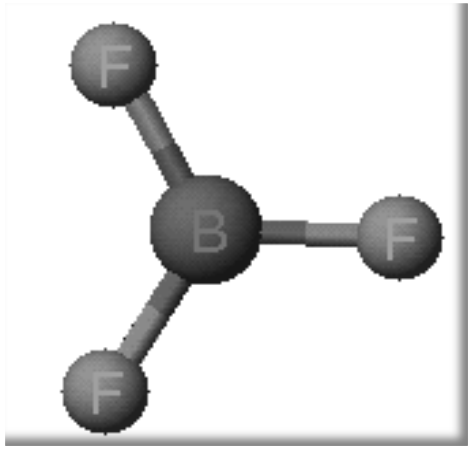
Bonding in BF_3

The atomic orbitals on the central B atom can't accommodate three bonds!

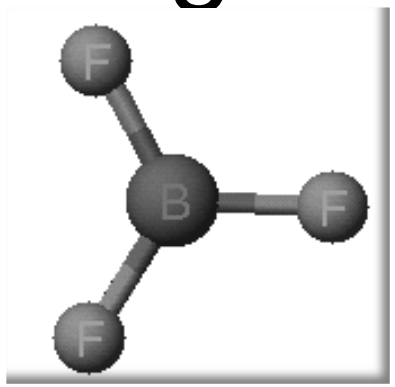


Boron configuration

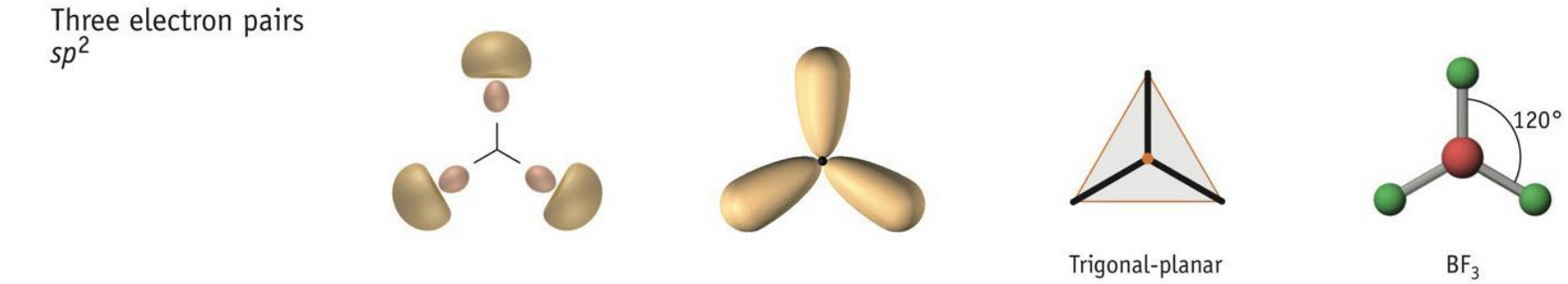




Bonding in BF_3



The 1 *s* orbital and 2 *p* orbitals must mix to form 3 new sp^2 hybrid orbitals.



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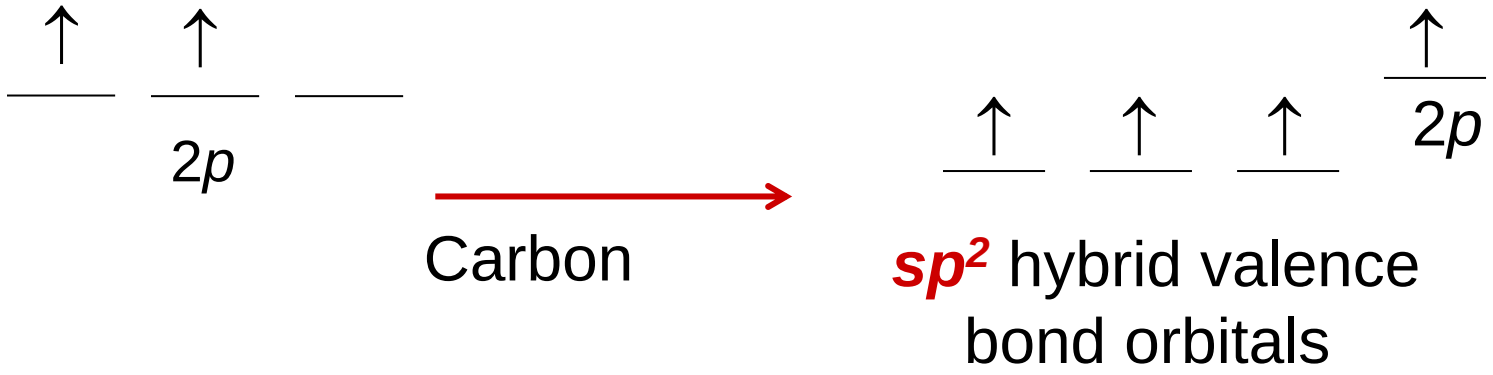
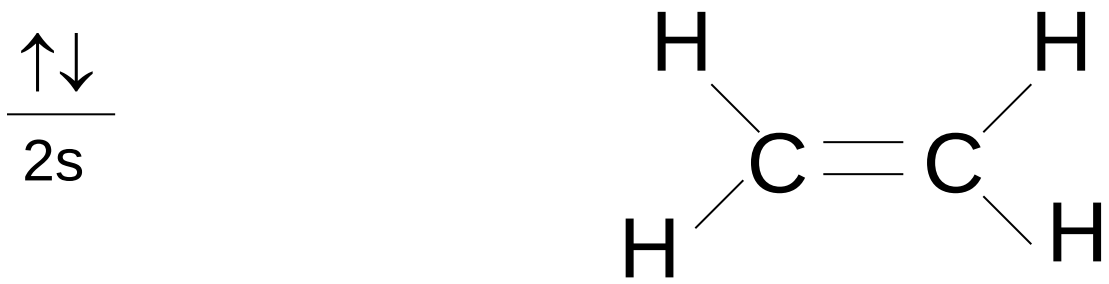
The 3 sp^2 hybrid orbitals can now form sigma bonds with each half-filled *p* orbital on each fluorine atom.
This results in a "*Trigonal-Planar*" molecular and electron-pair geometry.

sp^2 Hybridization

Another example of sp^2 hybridization is given by C_2H_4 (ethene)

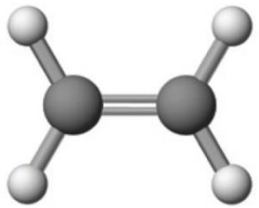




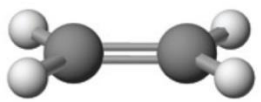
Trigonal-planar EPG at each carbon: sp^2 hybridization!

The left over p orbitals on each carbon overlap to form the π bond (second half of the double bond).
Each sp^2 orbital can form a σ bond, two with each of the H's and one with the other carbon.

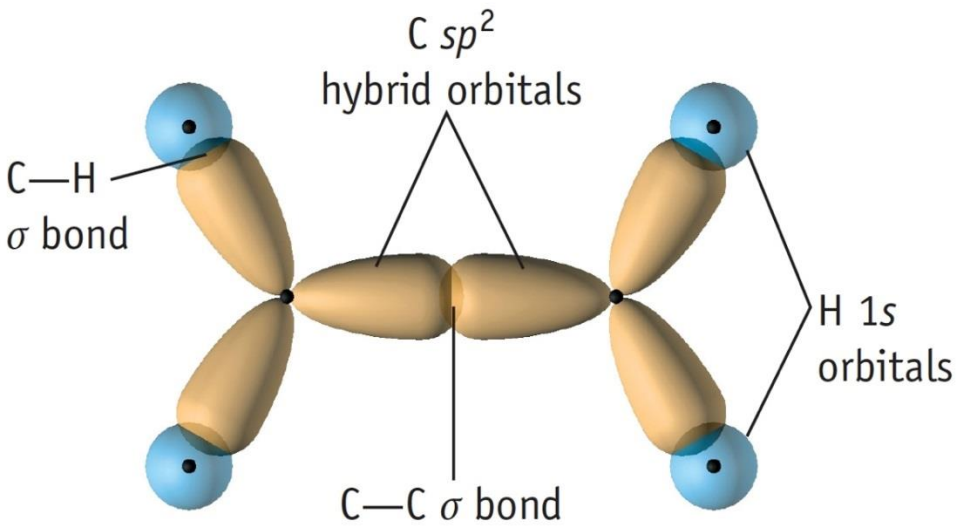
An Example of sp^2 Hybridization: C_2H_4

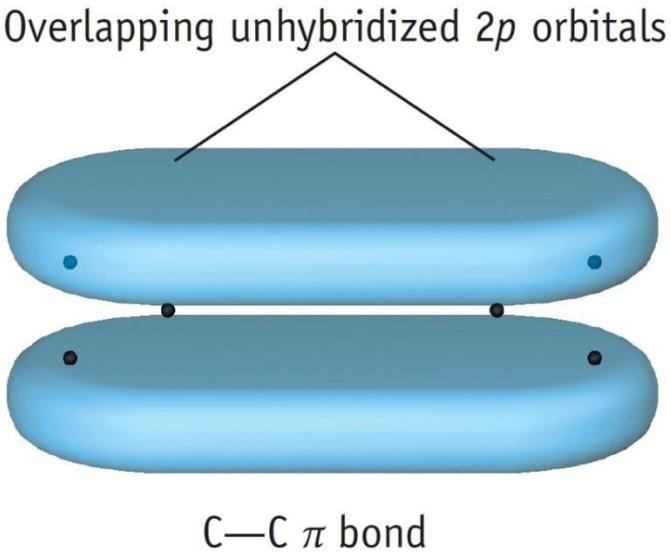


Top view



Almost side view



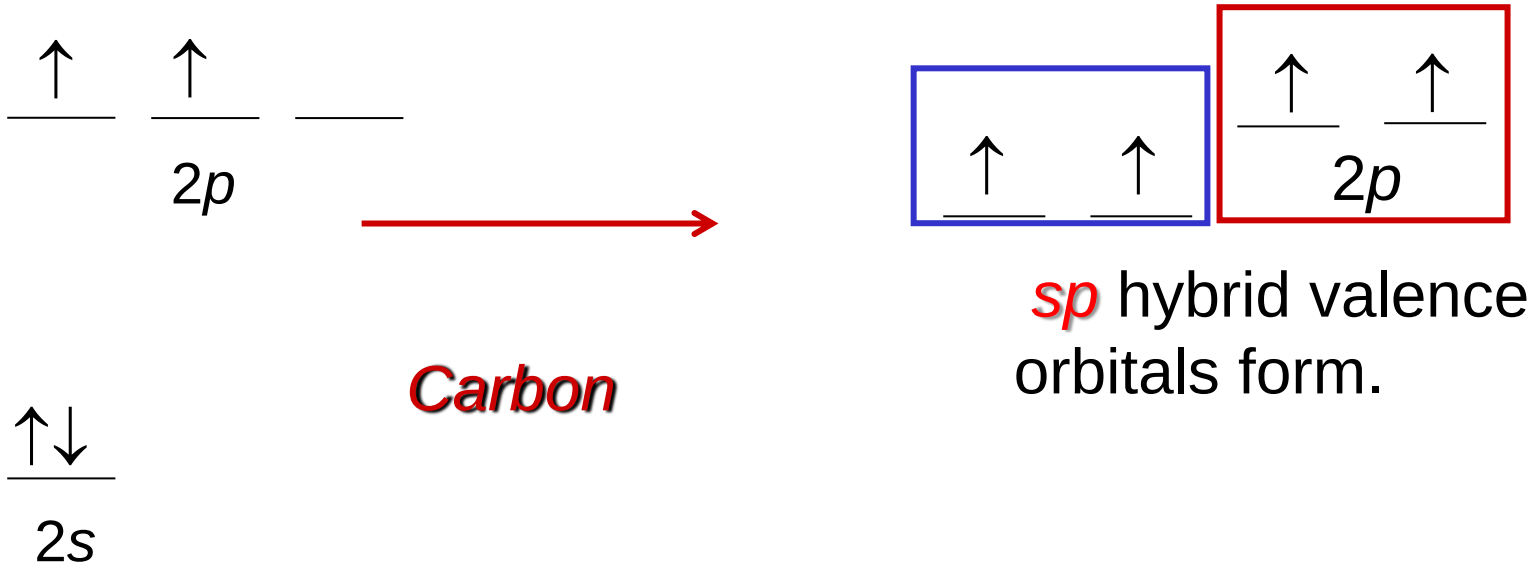


sp Hybridization

The *sp* orbitals spread out to form a *linear* geometry (directed away from one another) leaving the p orbitals perpendicular to the molecular axis.

The *sp* orbitals can form *σ bonds* or hold lone pairs. The two p orbitals can form the (2) *π bonds* in a triple bond.

An example of sp hybridization is given by C_2H_2 (acetylene*)*

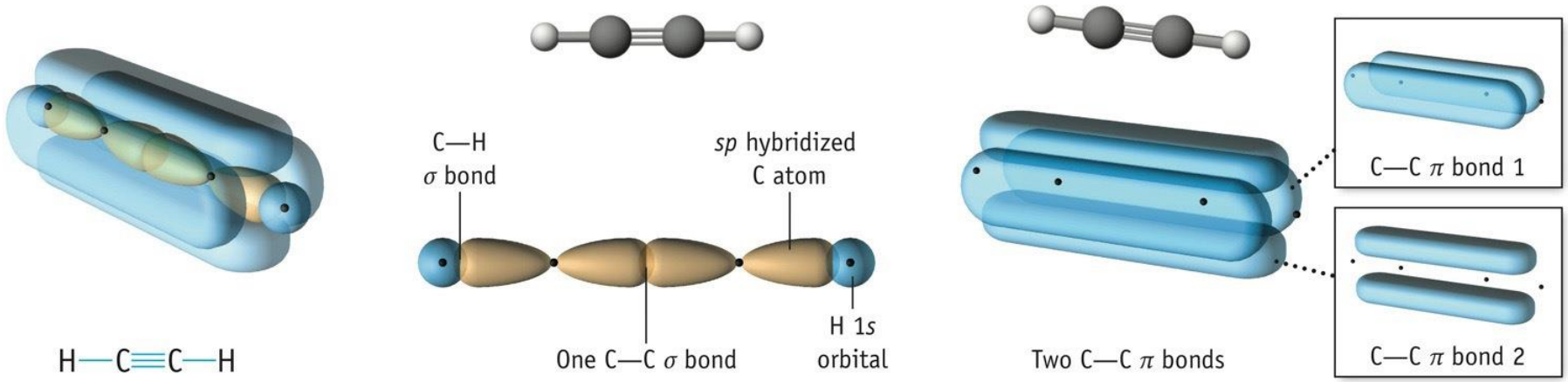


sp hybrid valence bond orbitals form.

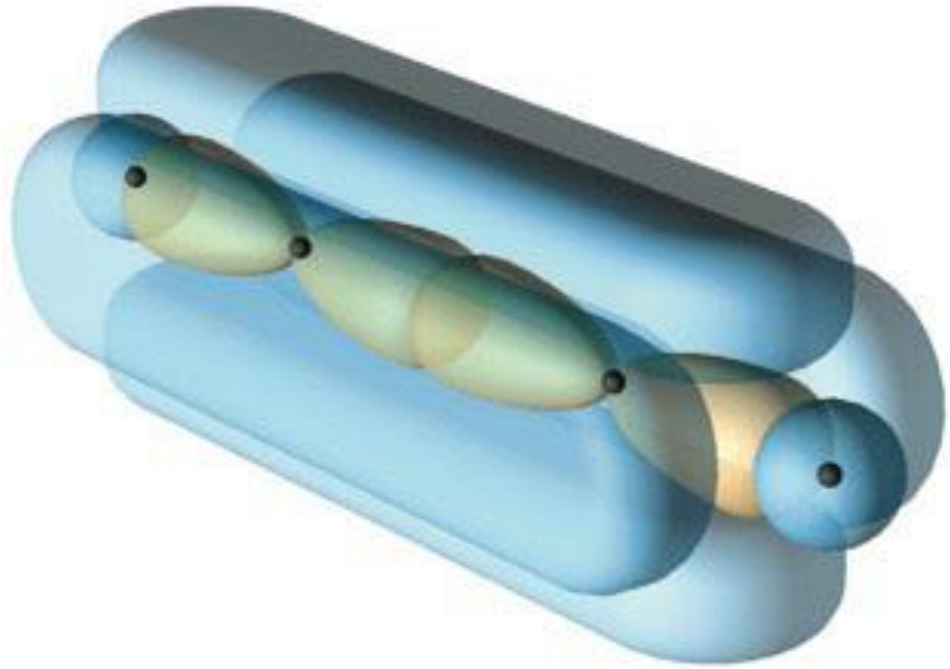
sp Hybridization

Just as with sp^2 hybridization, in *sp* hybridization, the left over *p* orbitals can form π bonds (in this case 2).

In acetylene (C_2H_2), there is a triple bond. (1 σ, 2 π's)



σ and π Bonding in C_2H_2



sp Hybridization

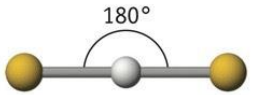
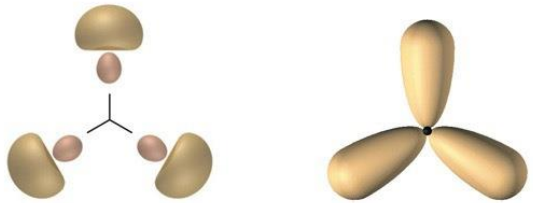
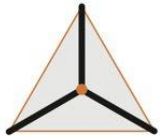
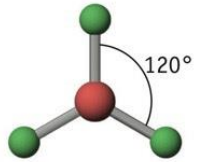
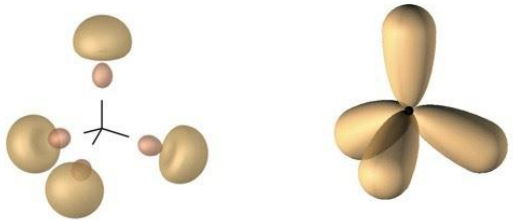
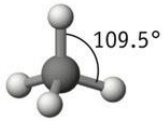
Other examples of molecules with sp hybridization are:

N_2	CN^- (cyanide anion)
:N≡N:	**[:C≡N:]$^-$**

<u>Conclusion:</u> When a central atom has a *linear* electronic geometry (EPG) with no lone pairs, it is most likely to bond through *sp* hybridization.

Compounds containing triple bonds (σ + 2π) or adjacent double bonds (CO_2) have *sp* hybridization.

sp, *sp*^{2}, & *sp*^{3} Hybridization

Hybrid Orbitals	Arrangement of Hybrid Orbitals	Geometry	Example
Two electron pairs *sp*		 Linear	 $BeCl_2$
Three electron pairs *sp*^{2}		 Trigonal-planar	 BF_3
Four electron pairs *sp*^{3}		 Tetrahedral	 CH_4

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Valence Bond Theory (2):Expanded Valence

For elements beyond the second period, we found several examples where the central atom in a Lewis structure had *greater than eight electrons* in the valence shell.

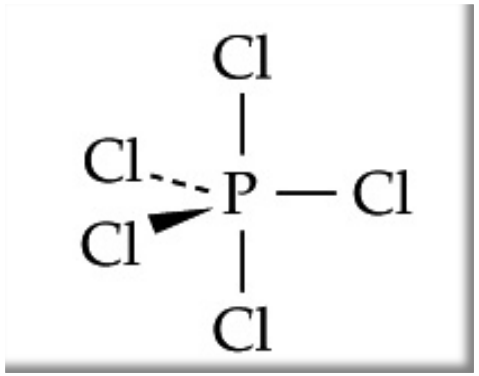
At n = 3	*l* =	0…	*s* orbitals
		1…	*p* orbitals
		2…	*d* orbitals

We've seen that when four sp^3 hybrid orbitals form, a central atom can accommodate only up to *eight electrons* in the valence (bp & lp).

To place more electrons in the valence, we must bring the *d orbitals* into the "mix."

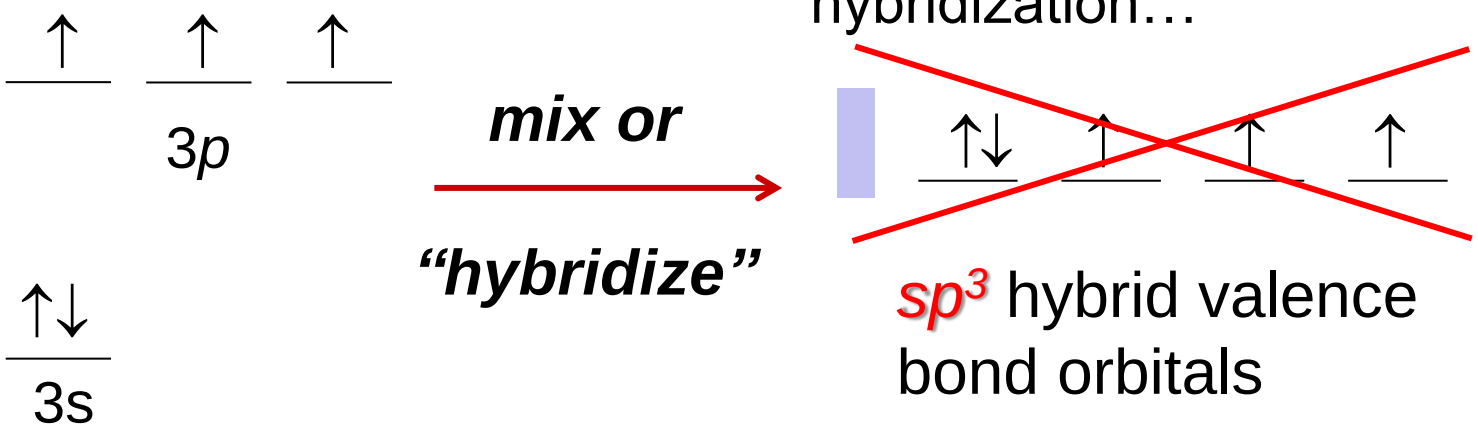
Consider a Molecule Like PCl_5

Phosphorous: valence electron configuration of $3s^2 3p^3$ *(five electrons)*
Each of the five electrons forms a single bond with a chlorine atom.



This means that central atom in the molecule needs five bonding orbitals to achieve the ***trigonal-bipyramidal*** electronic geometry.

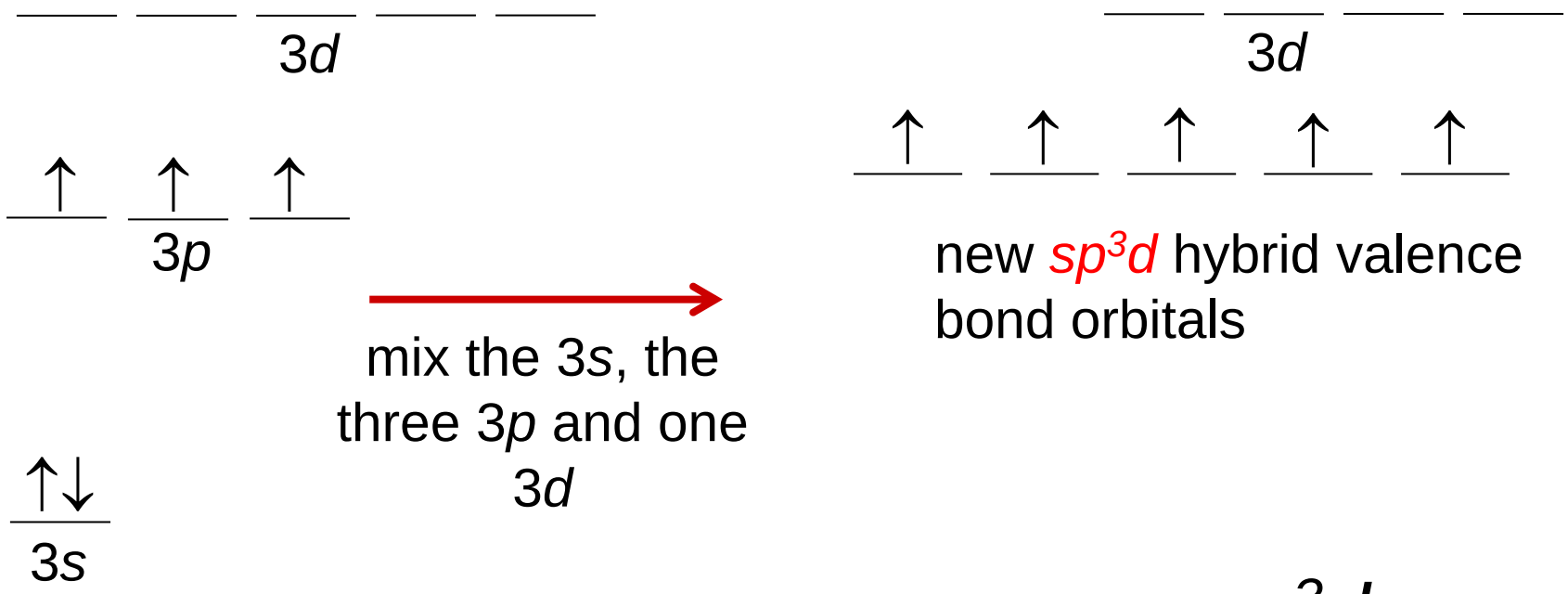
This cannot happen with sp^3 hybridization…



sp^3 hybrid valence bond orbitals

Valence Bond Theory (2): Expanded Valence

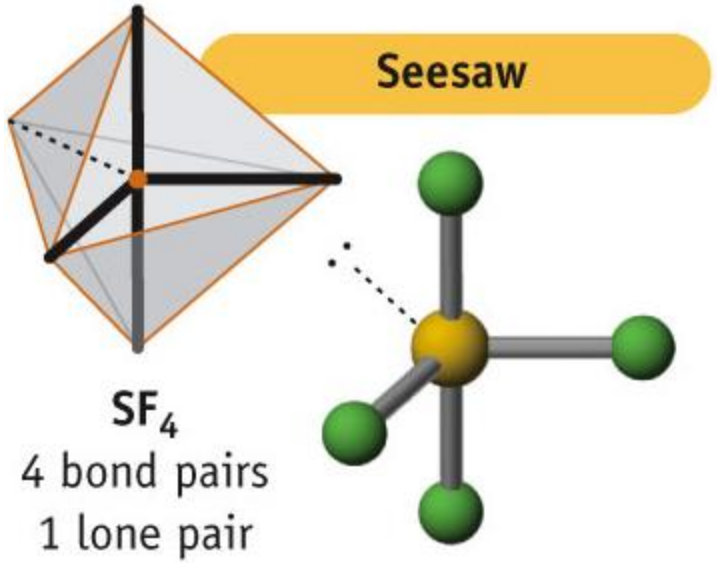
The only way to produce five half-filled orbitals on phosphorous is by adding a fifth atomic orbital…



Each of these half-filled sp^3d orbitals can form a σ bond with a chlorine atom in PCl_5.

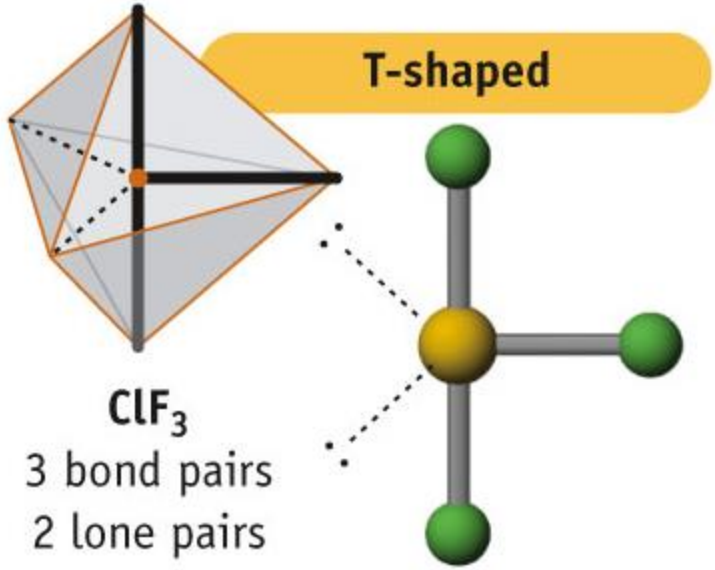
Additional Example of sp^3d Hybrid Molecules

SF_4 (sulfur tetrafluoride)



EPG: *Trigonal Bipyramidal*
MG: Seesaw

ClF_3 (chlorine trifluoride)

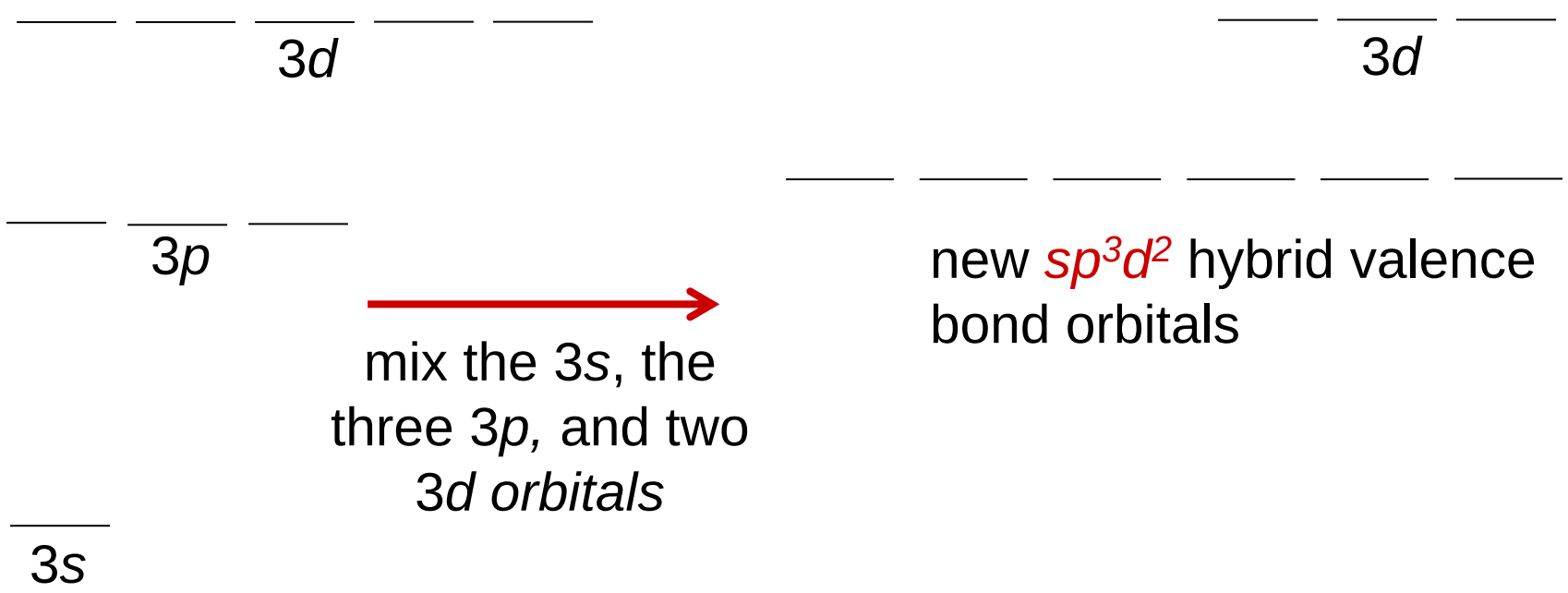


EPG: *Trigonal Bipyramidal*
MG: T-shape

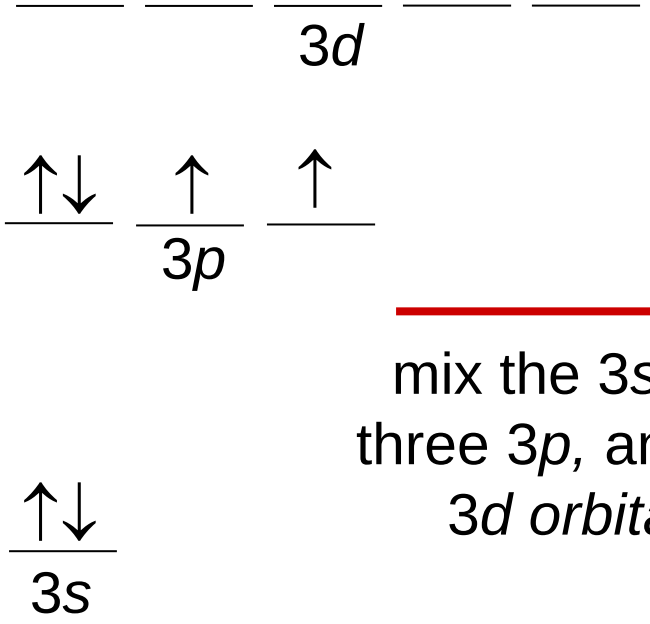
Valence Bond Theory (2): Expanded Valence

What about molecules with 12 electrons in the valence?

In order to achieve an expanded valence that can hold six electron pairs (bp & lp), we need to form six new hybrid orbitals. This requires the mixing of an s, three p's and two d atomic orbitals.

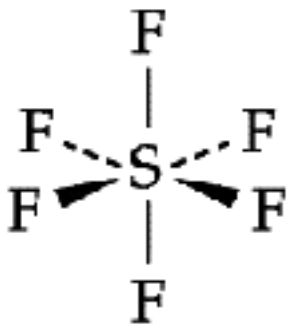


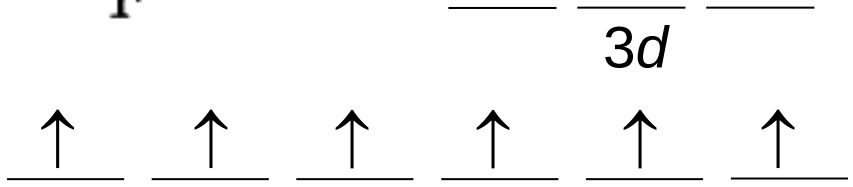
sp^3d^2 Hybridization: SF_6



sulfur

mix the 3*s*, the three 3*p*, and two 3*d orbitals*





six new sp^3d^2 hybrid valence bond orbitals

Each of these half-filled sp^3d^2 orbitals can form a σ bond with a fluorine atom.

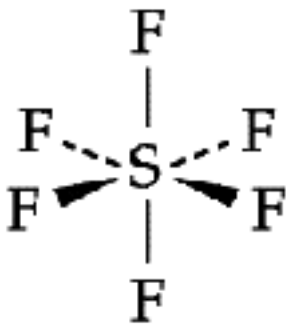
sp^3d^2 Hybridization: SF_6

Sulfur has a valence electron configuration of $3s^2 3p^4$ (six electrons).

Each of the six electrons forms a single bond with a fluorine atom forming an *octahedral* MG and EPG.

The bonding can be described in terms of sp^3d^2 hybrid orbitals.

SF_6



Predicting Hybridization

1. Start by drawing the Lewis structure (check formal charges).
2. Use VSEPR theory to determine the electron group geometry about the central atom.
3. Relate the central atom electron group geometry to the corresponding hybridization.
4. Identify and label the orbital overlap in each bond.
5. Label the bonds with σ and π bonds.

Molecular Orbital Theory

- Molecular Orbital Theory has several advantages and differences over VSEPR & VB theory:
- MO does a good job of predicting electron pair spectra and *paramagnetism*, where *VSEPR* and the VB theories don't.
- MO theory like VB theory predicts the bond order of molecules; however, it does not need resonance structures to describe molecules.
- The main drawback to our discussion of MO theory is that we are limited to talking about *diatomic molecules* (molecules that have only two atoms bonded together), or the theory gets very complex.

The Molecular Orbital Model (MO)

- Consider H_2
- Draw the Lewis structure
- What is the electron configuration of Hydrogen?
- Which valence electrons are used?
- Which orbital overlap is used?
- MOs are made up of individual AOs (Atomic Orbitals)

MO Theory

- $1s_a$ and $1s_b$ represent the $1s$ orbital from the 2 separated hydrogen atoms.
- $MO_1 = 1s_a + 1s_b$
- $MO_2 = 1s_a - 1s_b$

- When orbitals are added, the matching phases produce constructive interference, which gives enhanced electron probability between the nuclei.
  - This is a bonding MO.

- When one orbital is subtracted from the other, destructive interference occurs between the opposite phase, leading to a node between the nuclei.
  - This is an antibonding MO.

MO Theory

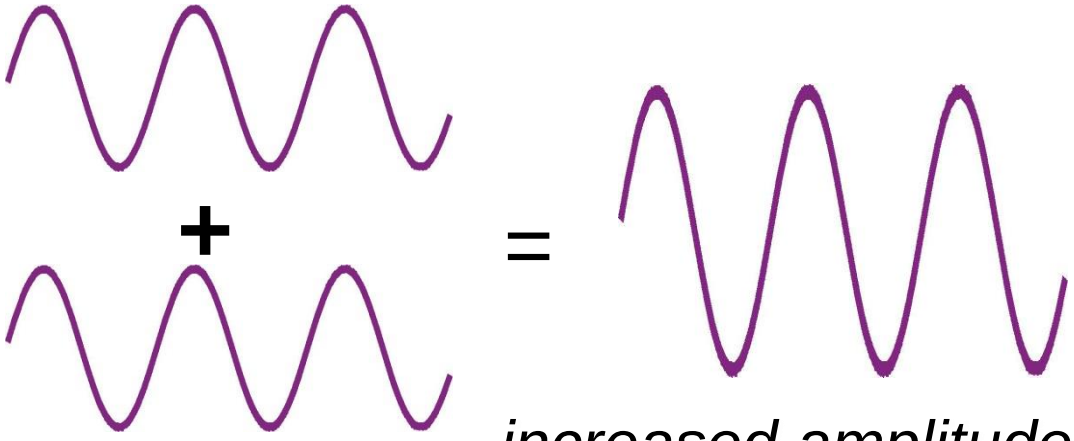
<u>MO Theory: *Consider Hydrogen*</u>

When two *wave functions* (orbitals) on different atoms add constructively, they produce a new MO that promotes bonding given by:

atomic orbitals new *Molecular Orbital*

$$\Psi(1s)_{H(1)} + \Psi(1s)_{H(2)} \rightarrow \Omega_{H\text{-}H}$$

when two waves add,
the amplitude increases:



"constructive interference"

increased amplitude

MO Theory

<u>MO Theory: *Consider Hydrogen*</u>

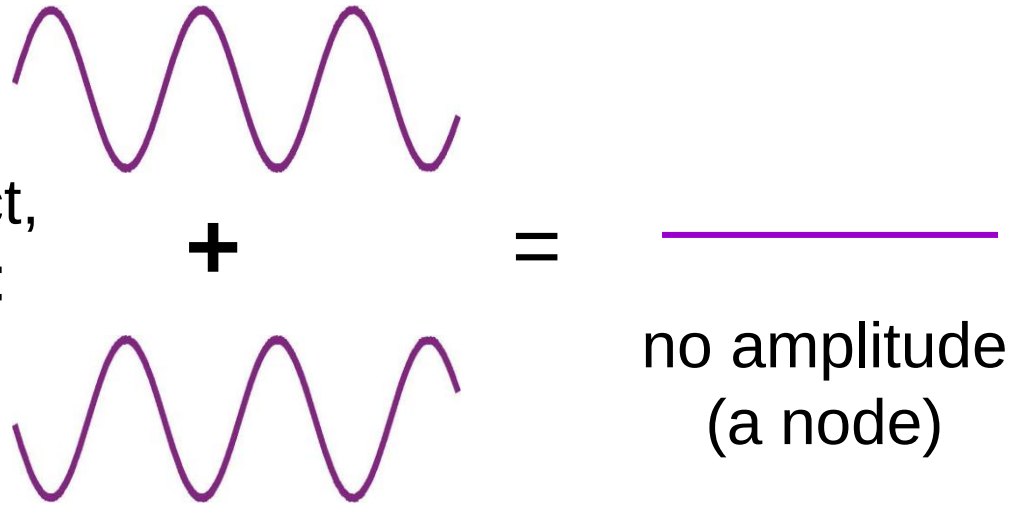
When two *wave functions* (orbitals) on different atoms add destructively, they produce a new MO that decreases bonding given by:

atomic orbitals new *Molecular Orbital*

$$\Psi(1s)_{H(1)} - \Psi(1s)_{H(2)} \rightarrow \Omega^{*}_{H\text{-}H}$$

When two waves subtract, the amplitude decreases:

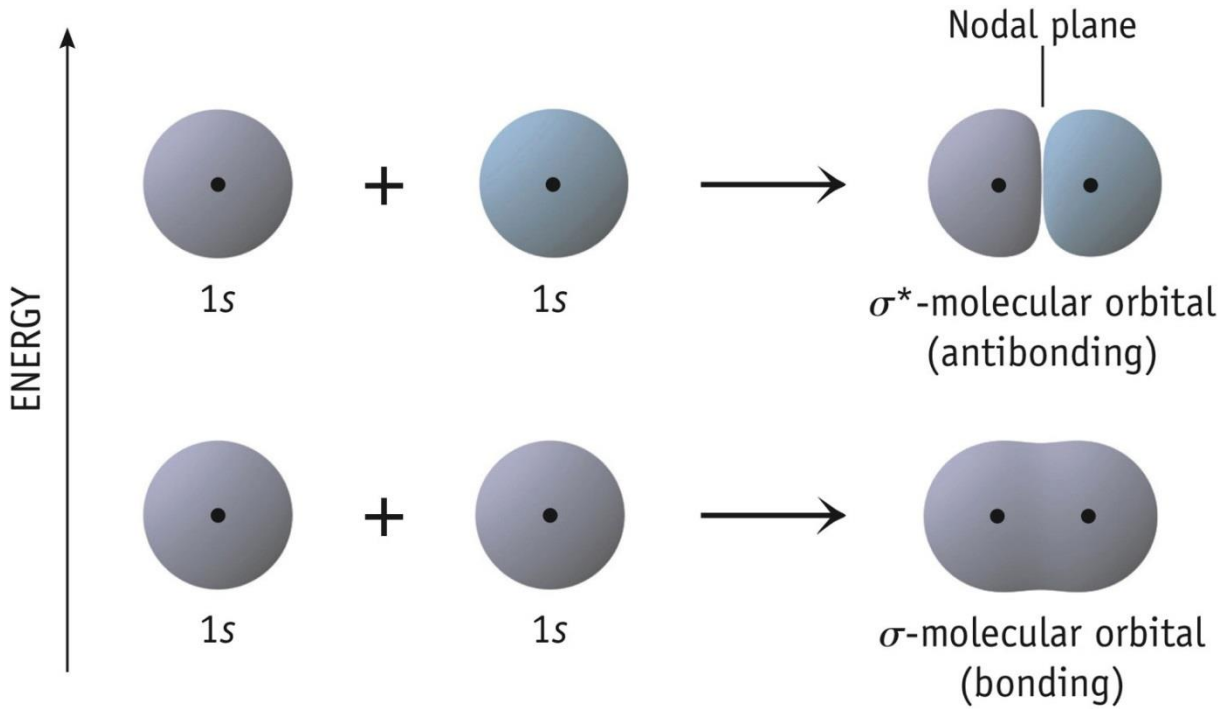
"destructive interference"



no amplitude
(a node)

Molecular Orbital Theory

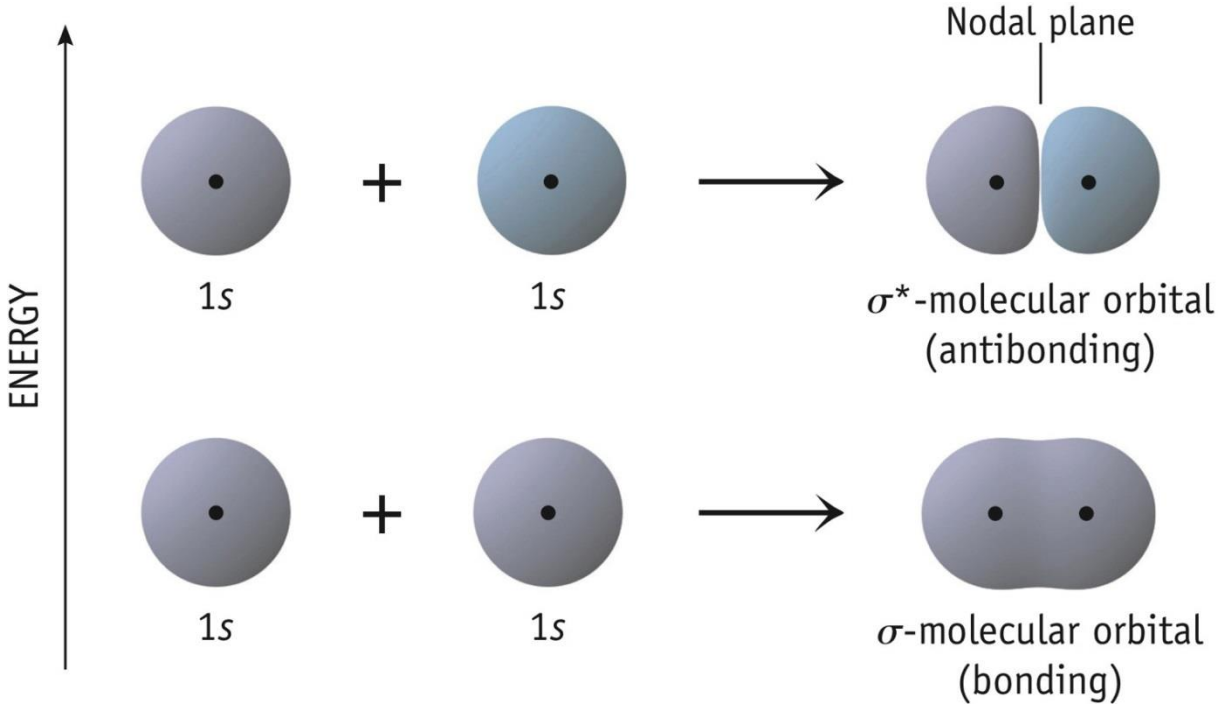
Consider hydrogen atoms combining to form H_2.
The individual 1*s* atomic orbitals combine.



When they add, a lower energy σ-bonding MO forms.
When they subtract, a higher energy σ*-antibonding MO forms.

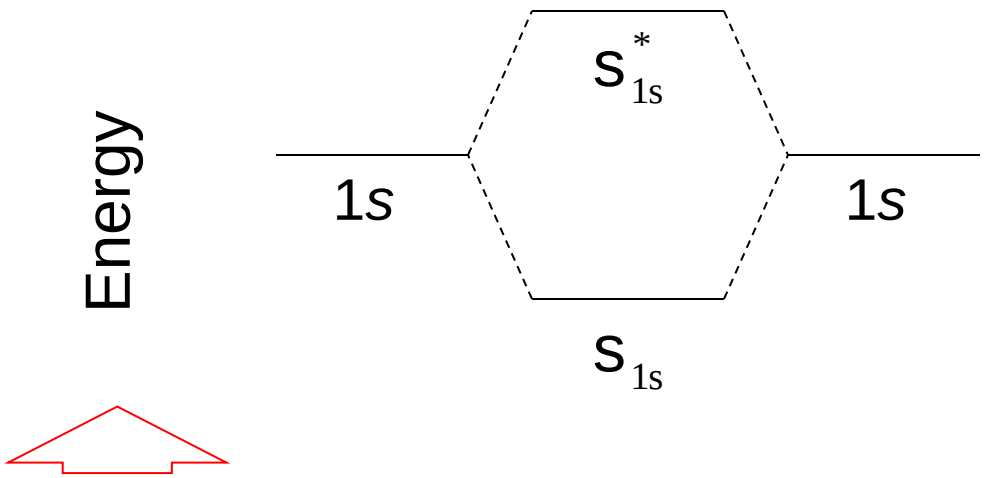
Molecular Orbital Theory

- Bonding and antibonding sigma MOs are formed from 1*s* orbitals on adjacent atoms.



Molecular Orbital Diagrams

Molecular orbitals result when atomic orbitals on bonding atoms constructively and destructively combine:

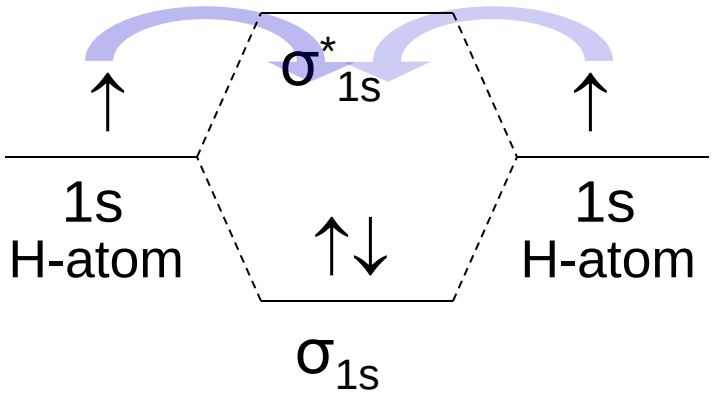


When they add, lower energy bonding Molecular Orbitals (MOs) form; when they subtract, higher energy antibonding MOs form.

Antibonding orbitals are designated by an asterisk (*) called a *star*.

Molecular Orbital Diagrams

Once again, we consider the simplest molecule, H_2.
When two hydrogen atoms combine, the 1s orbitals on each can add or subtract to form a σ_{1s} bonding or σ^*_{1s} antibonding orbital.



Each H atom has one *1s* electron that can contribute to the MO bonding and antibonding MOs.

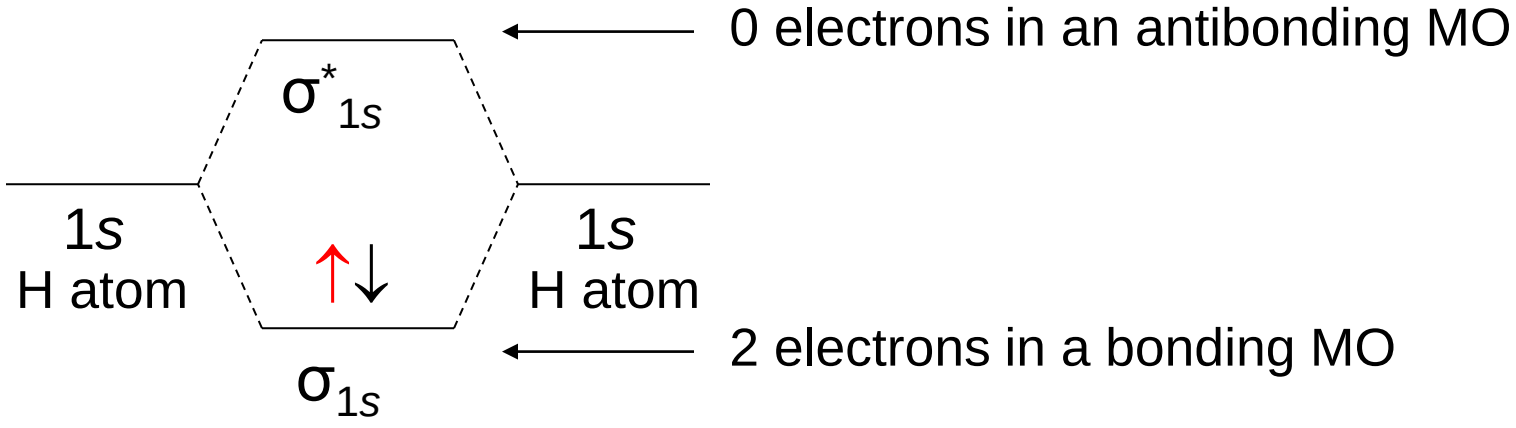
Just as in the electron configurations of atoms, the electrons fill from the lowest energy MO first (*Aufbau principle*) only pairing when forced to (*Hund's rule*). Each MO can only hold two electrons of opposite spin (*Pauli principle*).

Bond Order in Molecular Orbital

$$\text{BO} = \frac{\text{\# of Bonding electrons} - \text{\# antibonding electrons}}{2}$$

- Electrons in bonding molecular orbitals add stability.
- Electrons in antibonding molecular orbitals reduce stability.

Bond Order: The bonding in H_2 is described by…



$$\text{BO} = \frac{\text{\# of Bonding electrons} - \text{\# antibonding electrons}}{2} = \frac{2 - 0}{2} = 1$$

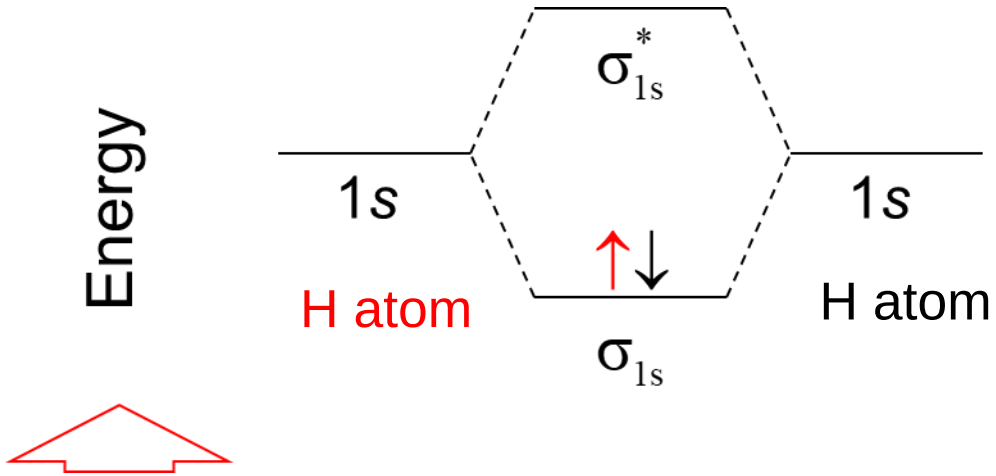
Bond Order = 1 (single bond)

MO electron configuration: $(\sigma_{1s})^2$

One also sees that the molecule (H_2) is ***diamagnetic*** (no unpaired electrons).

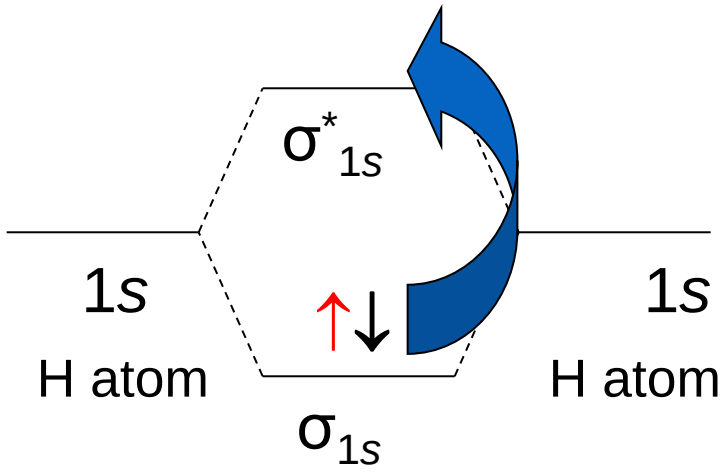
MO Energy Diagrams for H_2

- Bonding will result if the molecule has lower energy than the separated atoms.
- A very stable molecule of H_2 is lower in energy than the separated hydrogen atoms by 432 kJ/mol.



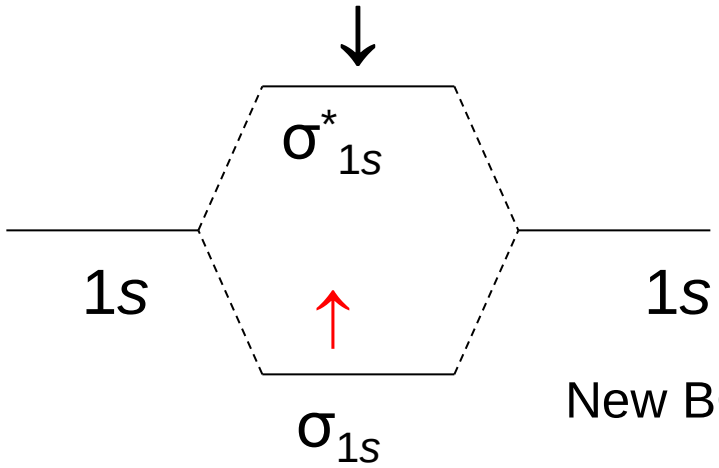
Excited States

What happens to a H_2 molecule if one of the electrons is excited to the antibonding orbital?



$$\text{Original BO} = \frac{2 - 0}{2} = 1$$

ground state H_2



$$\text{New BO} = \frac{1 - 1}{2} = 0$$

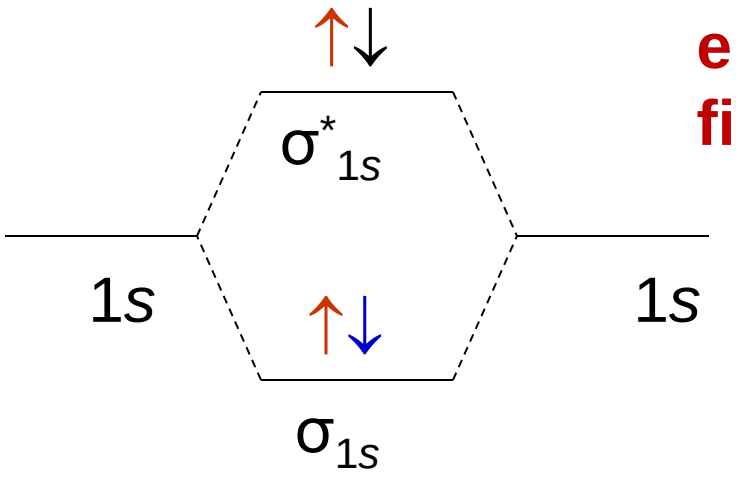
excited state H_2

The molecule falls apart!

Photodissociation

Why Doesn't He_2 Exist?

He: $1s^2$

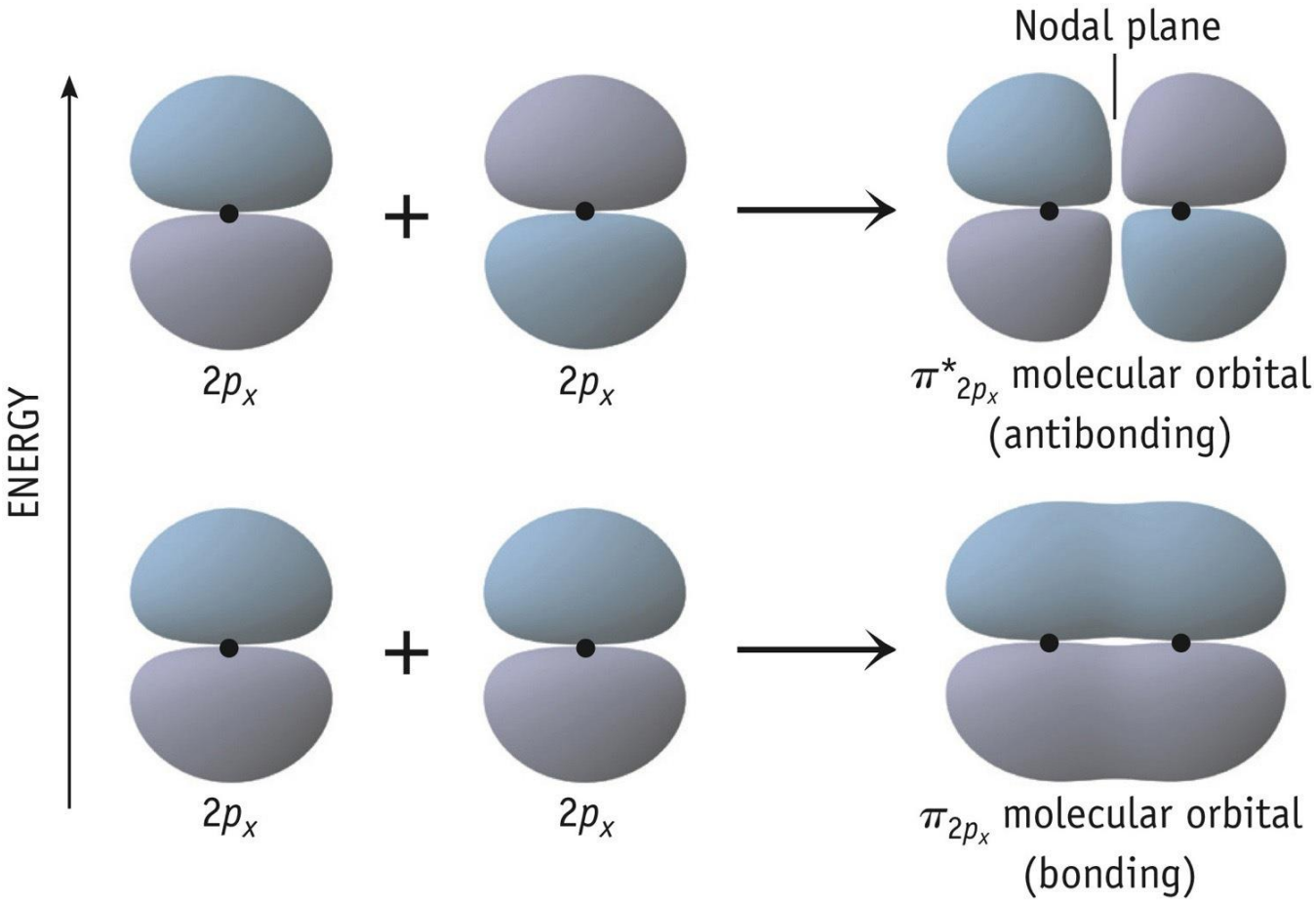


electrons in AOs fill the MOs

$$BO = \frac{2 - 2}{2} = 0$$

Molecular Orbital theory predicts that the molecule is unstable!

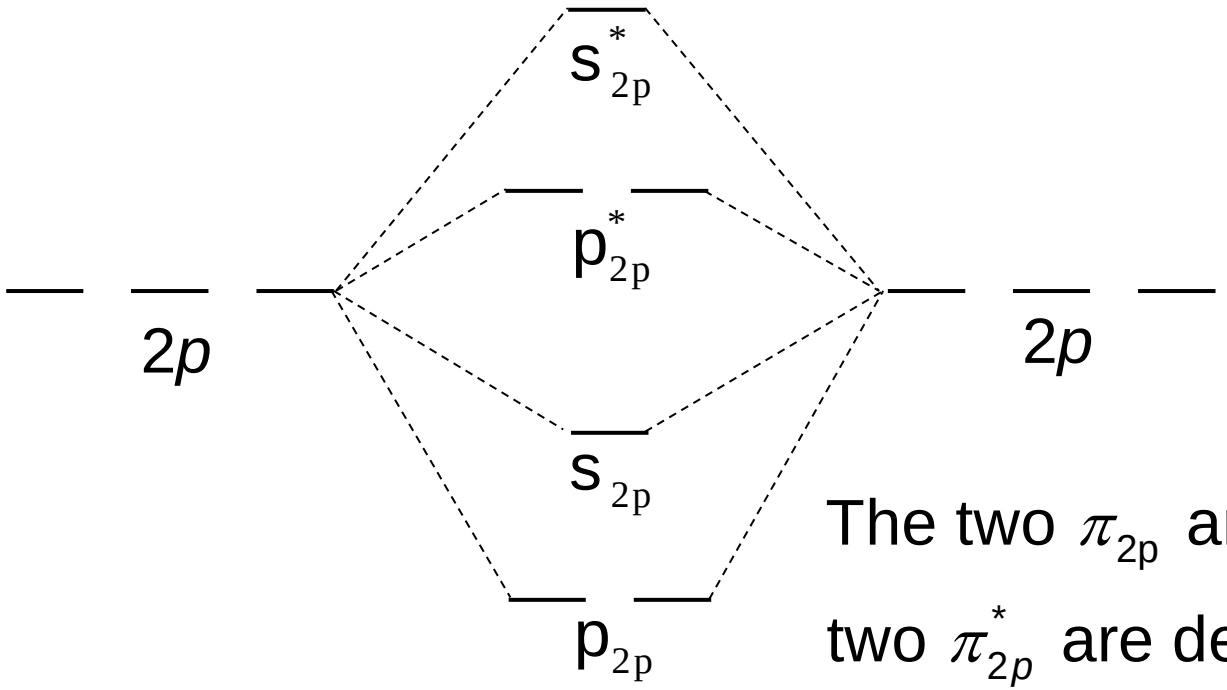
Molecular Orbitals from Atomic *p* Orbitals



Sideways overlap of atomic 2*p* orbitals that lie in the same direction in space give π bonding and antibonding MOs.

The MO Correlation Diagram for the 2*p* Atomic Orbitals

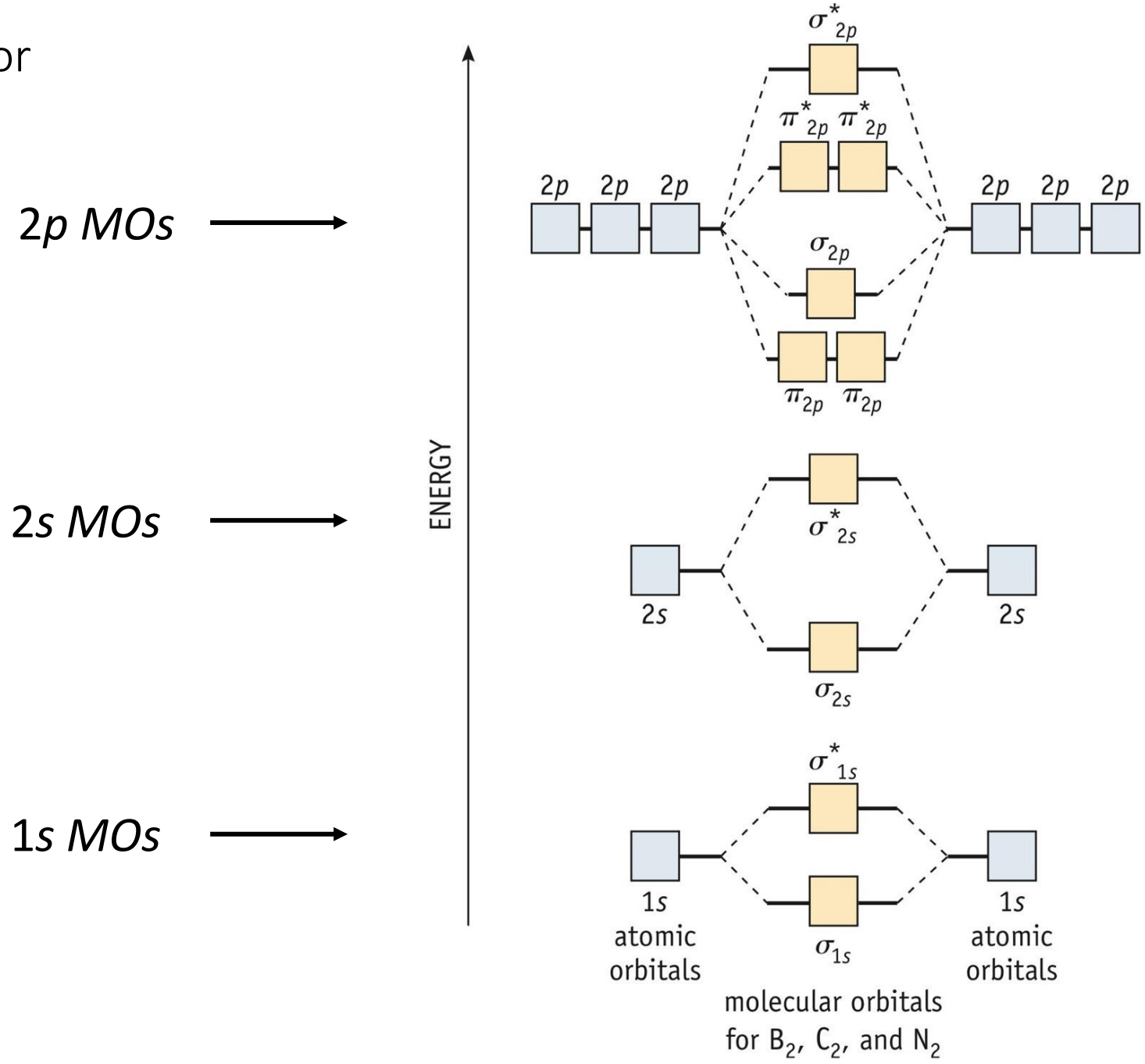
The three 2*p* orbitals on each side combine to form six new Molecular Orbitals that can accommodate up to 12 electrons.



The two π_{2p} and the two π^*_{2p} are degenerate.

This configuration is seen for B, C, and N only…

The overall MO diagram for the 1*s*, 2*s*, and 2*p* orbitals



The MO Correlation Diagram for the 2*p* Atomic Orbitals

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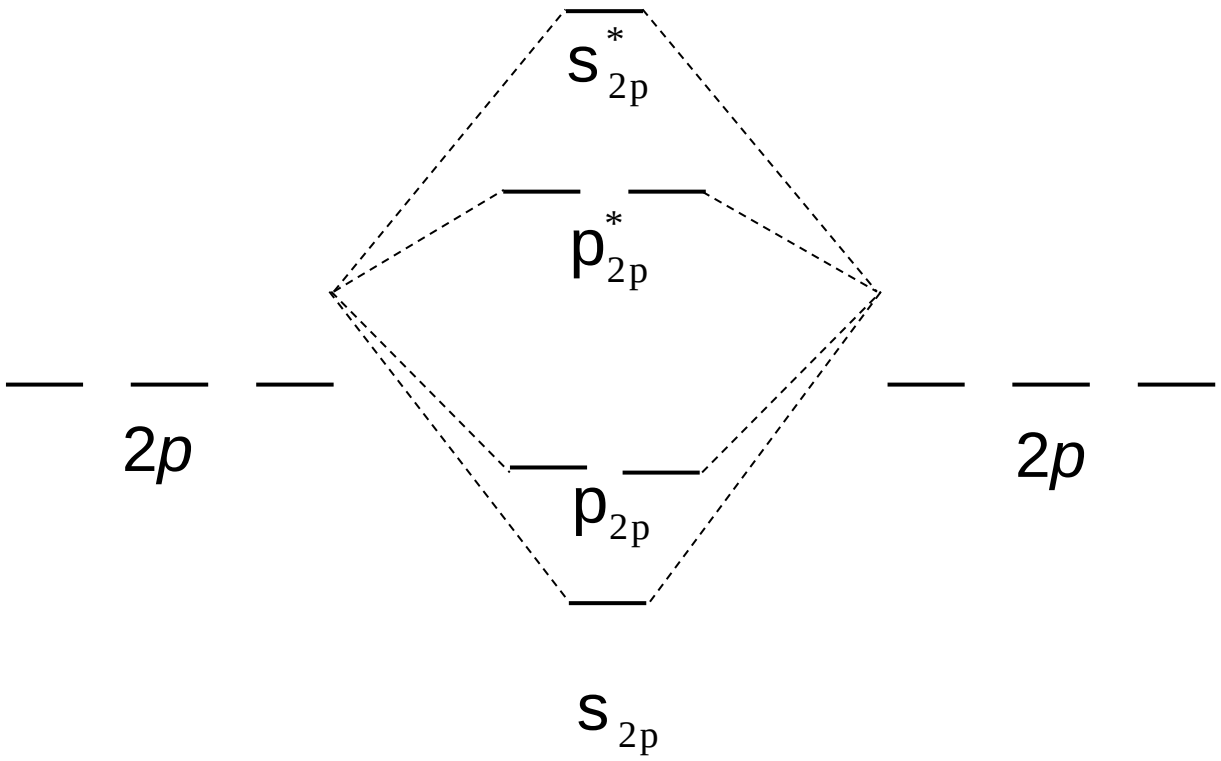
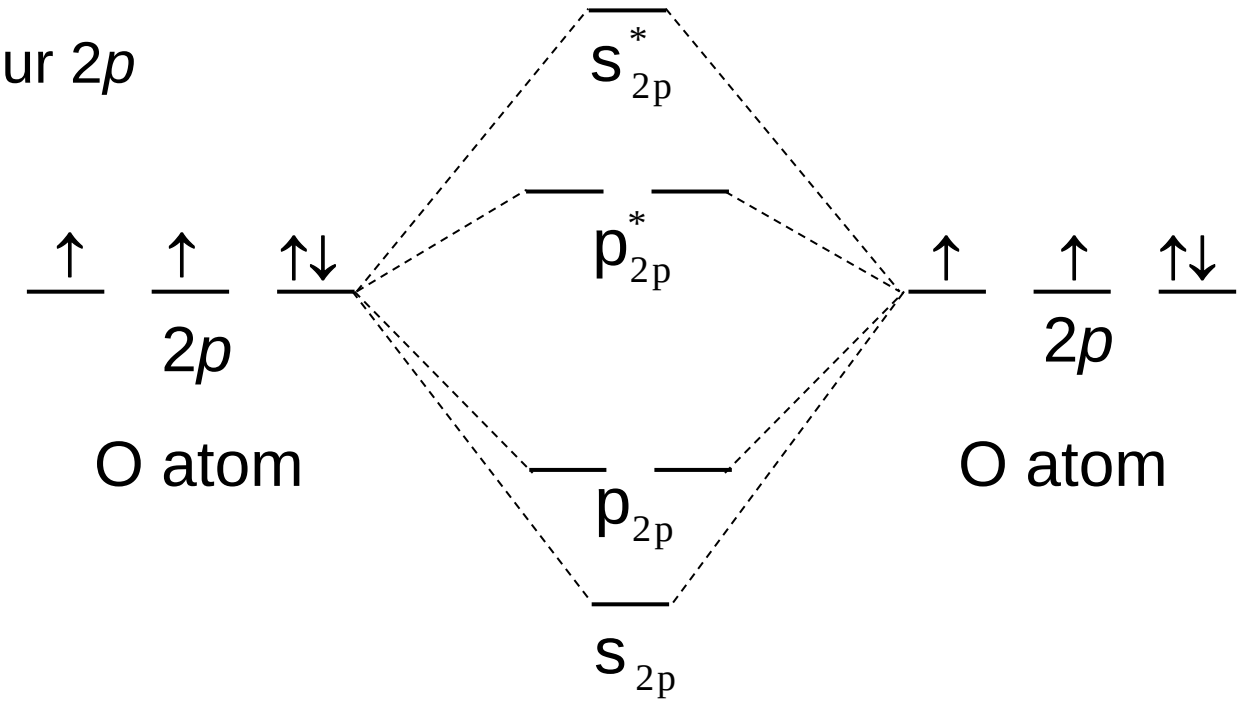


This configuration is seen for O and F only…

Bonding in O_2

Since the 1*s* and 2*s* MOs are full, they do not contribute to any net bonding. Any net bonding will be determined by the 2*p* MOs.

Each O atom has four 2*p* electrons:



The electrons fill the MOs by the Aufbau principle and Hund's rule.

Molecular Orbital Occupations and Physical Data for Homonuclear Diatomic Molecules of Second-Period Elements

TABLE 9.1 Molecular Orbital Occupations and Physical Data for Homonuclear Diatomic Molecules of Second-Period Elements*

	B_2	C_2	N_2		O_2	F_2
σ^*_{2p}				σ^*_{2p}		
π^*_{2p}				π^*_{2p}	↑ ↑	↑↓ ↑↓
σ_{2p}			↑↓	π_{2p}	↑↓ ↑↓	↑↓ ↑↓
π_{2p}	↑ ↑	↑↓ ↑↓	↑↓ ↑↓	σ_{2p}	↑↓	↑↓
σ^*_{2s}	↑↓	↑↓	↑↓	σ^*_{2s}	↑↓	↑↓
σ_{2s}	↑↓	↑↓	↑↓	σ_{2s}	↑↓	↑↓
Bond order	One	Two	Three		Two	One
Bond-dissociation energy (kJ/mol)	290	620	945		498	155
Bond distance (pm)	159	131	110		121	143
Observed magnetic behavior (paramagnetic or diamagnetic)	Para	Dia	Dia		Para	Dia

Bonding in O_2

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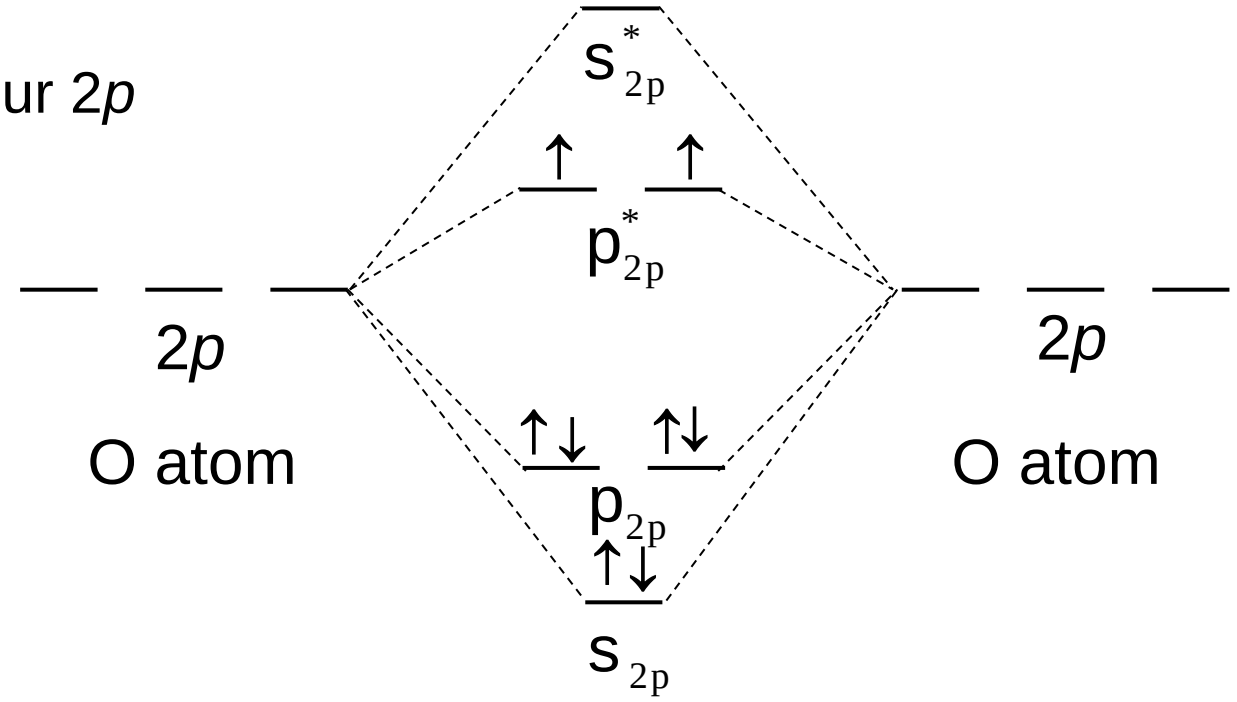


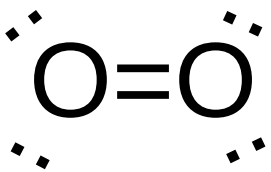
The electrons fill the MOs by the Aufbau principle and Hund's rule.

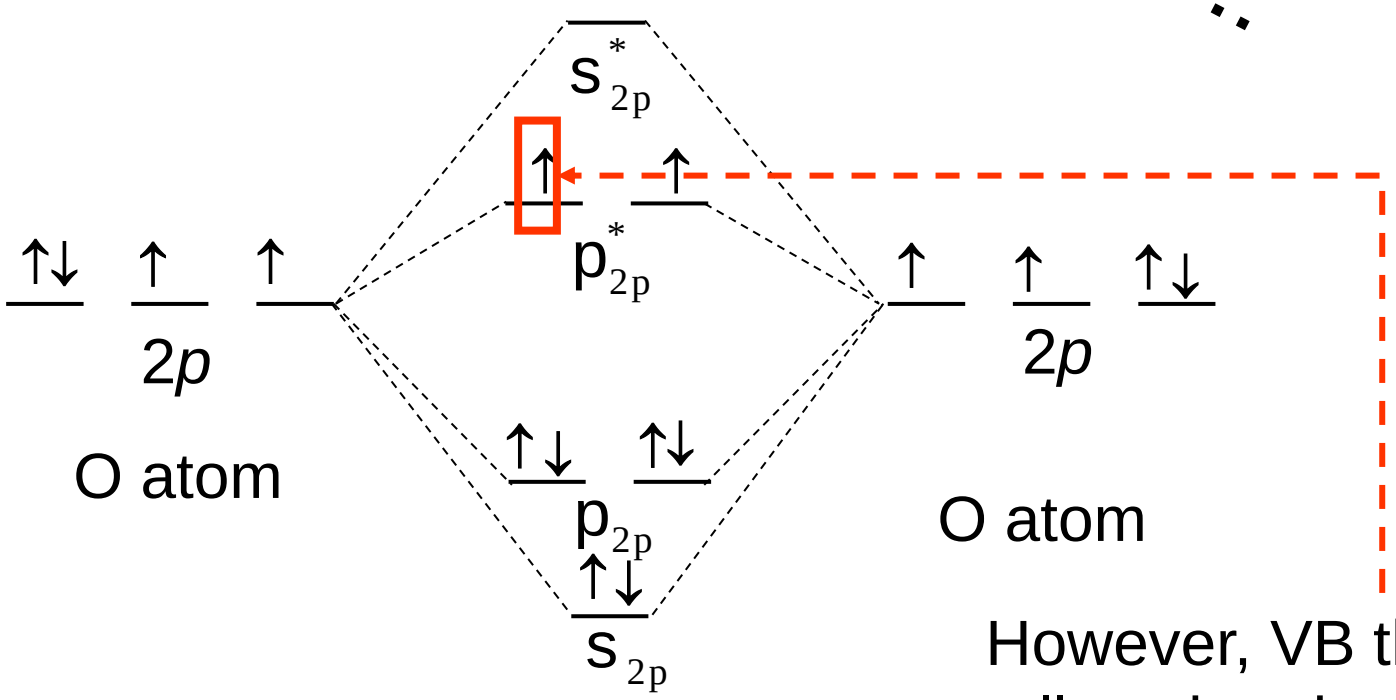
Bonding in O_2

https://www.youtube.com/watch?v=PG2w8d4msM4

$$BO = \frac{6 - 2}{2} = 2$$ ***double bond***

This agrees with the Lewis dot structure:





However, VB theory did not tell us that the molecule is paramagnetic!

Bond Order

- In Molecular Orbital Theory, the stability of a covalent bond is related to its bond order.
- *A bond order of 1 represents a single bond.*
- *A bond order of 2 represents a double bond.*
- *A bond order of 3 represents a triple bond.*
  - Because MO theory also treats molecules with an odd number of electrons, bond orders of ½, 3/2, or 5/2 are possible.

Bond Order

- Larger bond order means larger strength

$$\text{Bond order} = \frac{\text{number of bonding electrons} - \text{number of antibonding electrons}}{2}$$

- Bond order of H_2 = $\frac{2 - 0}{2} = 1$
- Bond order of H_2^- ion = $\frac{2 - 1}{2} = 1/2$